Andreu Marqués Goyes

Backend / Systems Engineer | Secure Runtime Execution, High-Concurrency APIs & Data Architecture

Palma de Mallorca, Spain | +34 640 142 281 | andreuvaleria25@gmail.com

linkedin.com/in/andreuvm | github.com/AndreuVM | *Available for On-site or Remote Roles*

Education

Universidad Alfonso X el Sabio (UAX) — Madrid / Remote, Spain

Higher Vocational Degree in Multiplatform Application Development (DAM) — *2025 – Present*

Specialized in enterprise back-end development, relational databases, concurrent programming, and AI pipelines.

Universitat de les Illes Balears (UIB) — Palma, Spain

B.S. in Computer Engineering (96 ECTS Completed – CS Foundations) — *2020 – 2024*

Rigorous university coursework in Computer Science: Algorithms & Data Structures I and II (Honor/Notable), Computer Programming I (Grade: 9.2/10), Operating Systems I, Computer Architecture I and II, Software Engineering, and Applied Mathematics (Calculus, Statistics, Linear Algebra).

Technical Skills

Languages: C, C++, Java, Python, SQL (PostgreSQL, MySQL, Oracle SQL), Bash/Shell, Dart/Flutter, Assembly (68k)
Systems & Observability: Linux/Unix Internals, Memory Management, Process Scheduling, POSIX, Docker, Git, Telemetry
Back-End & Cloud: RESTful APIs, FastAPI, Microservices, Spring Boot, Asynchronous I/O, CI/CD, Multithreading / Concurrency
Databases & Storage: Schema Architecture, Query Optimization, Transactional ACID, Indexing (B-Trees)
AI-Assisted Development: Agentic Workflows, MCP (Model Context Protocol) Server Development, LLM Tooling

Featured Technical Projects

PRAXEON: Runtime Supervision & Security for AI Agents | github.com/AndreuVM/praxeon

Python, Docker, MCP, HMAC Cryptography, FastAPI — *293 passing tests*

- Engineered a deterministic middleware decoupling stochastic LLM proposals from physical execution via time-bounded HMAC-SHA256 capabilities and durable anti-replay nonce storage.
- Implemented multi-tier sandboxing (OCI containers with `--read-only` and local jail mitigating symlink traversal/SSRF), native MCP server, and empirical benchmarks ($p50 \sim 0.22$ ms).

Custom Unix Shell & Virtual File System Simulator | GitHub

C/C++, Linux Systems Programming, POSIX, Valgrind

- Built a Unix-like shell from scratch featuring custom process lifecycle management (`fork`, `exec`, signals) and zero-leak memory verified with Valgrind.
- Implemented a virtual file system simulator with block-based disk allocation, inode pointers, path resolution, and multi-tier POSIX permission validations.

High-Performance Trie-Based Dictionary & Word Engine | GitHub

Java, Android SDK, Data Structures, Regex

- Designed an optimized prefix tree (Trie) data structure to perform $O(L)$ dictionary lookups across tens of thousands of words in memory.
- Built a high-throughput pattern matching module utilizing dynamic Regex compilation to validate anagram permutations with sub-millisecond execution latency.

Smart Restaurant Capacity & Queue Optimization Engine | GitHub

Python, FastAPI, Scikit-Learn, Docker, SQL

- Architected an asynchronous backend service with ML regression predicting table turn-times to optimize seat allocation, reducing customer wait times by 28%.
- Containerized services with Docker and engineered high-concurrency REST APIs with automated schema validation and structured JSON telemetry.

Experience

Independent Software Developer — Palma de Mallorca, Spain

Freelance Web & Backend Developer — *2020 – Present*

- Engineered and deployed custom software and web solutions for private clients, integrating modular architectures, database design, and Git version control.

Revomon — Remote

Freelance 3D Technical Artist & Asset Optimization — *Jun 2022 – Feb 2023*

- Optimized 3D digital assets under strict real-time memory and polygon constraints, developing automated validation scripts for technical pipelines.

Languages

English: Full Professional Proficiency | **Spanish:** Native | **Catalan:** Native | **Portuguese:** Intermediate